HIERO-CODE · SYMBOLIC COMPUTE



The Reference Codex

A field guide to the rune & numeral systems — Egyptian, Sumerian, Mayan, Khoisan, and the Pyri emoji layer.



Hiero-Code treats ancient scripts as computational alphabets. Each system below is a different way of packing meaning into a few repeating marks — base-10 sums, base-60 places, base-20 stacks, parallel click channels — and the Pyri system extends that same instinct from counting quantity to encoding emotion.

I·EGYPTIAN II·SUMERIAN III·MAYAN IV·KHOISAN
V·GLYPH LIBRARY VI·PYRI EMOJI VII·EMOJI LINEAGE
VIII·IÑUPIAQ IX·AZTEC X·STELAE CORPUS
XI·HINDU ICONOGRAPHY XII·NORTHWEST COAST ART XIII·INU-IT SYMBOLOGY

Egyptian Hieroglyphic

Base 10 · Additive

Ancient Egyptian decimal notation using pictographic symbols. Each power of ten has a distinct hieroglyph; numbers are written by repeating and summing symbols — there is no place value. Attested from c. 3000 BC on papyrus and carved in stone.

Power Symbols

GLYPH	VALUE	NAME	MEANING
	1	Stroke	Single unit, the tally mark
	10	Heel Bone (Hobble)	The first grouping, ten strokes
	100	Coil of Rope	The coil, one hundred
	1,000	Lotus Flower	Abundance, the lotus in bloom
	10,000	Bent Finger	Power of the pharaoh's hand
	100,000	Tadpole / Frog	The infinite multiplier
	1,000,000	Kneeling God (Heh)	The god of eternity, a million

Unit Fractions — Rhind Papyrus style, "parts" written above a number

GLYPH	FRACTION	DECIMAL
	1/2	0.5
	2/3	0.6667
	3/4	0.75
	1/3	0.3333
	1/4	0.25
	1/5	0.2
	1/7	0.1429
	1/10	0.1

Worked example. 1,234 = 1×  + 2×  + 3× + 4× — read as a simple sum of grouped pictographs.

II Sumerian Sexagesimal

Base 60 · Positional

The world's first place-value system, c. 3000 BC, pressed in cuneiform wedges on clay. Base 60 is divisible by 2, 3, 4, 5, 6, 10, 12, 15, 20 and 30 — which is why it still rules our clocks (60 seconds, 60 minutes) and circles (360°).

Cuneiform Numerals

GLYPH	VALUE	NAME
	1	Aš (One)
	2	Min (Two)
	3	Eš (Three)
	4	Limmu (Four)
	5	Ia (Five)
	6	Aš (Six)
	7	Imin (Seven)
	8	Ussu (Eight)
	9	Ilimmu (Nine)
	10	U (Ten)
	20	Niš (Twenty)
	30	Ušu (Thirty)
	40	Nimin (Forty)
	50	Ninnu (Fifty)
	60	Géš (Sixty)
	600	Géš-U (Six Hundred)
	3,600	Šár (3,600)
	36,000	Šár-U (36,000)
	216,000	Šár-Gal (Great Šár)

Igi Fractions — reciprocals of the "regular" divisors of 60

RECIPROCAL	FRACTION	DECIMAL
igi-2	1/2	0.5

RECIPROCAL	FRACTION	DECIMAL
igi-3	1/3	0.3333
igi-4	1/4	0.25
igi-5	1/5	0.2
igi-6	1/6	0.1667
igi-10	1/10	0.1
igi-12	1/12	0.0833
igi-15	1/15	0.0667
igi-20	1/20	0.05
igi-30	1/30	0.0333
igi-60	1/60	0.0167

Worked example. 3,661 = 1×3600 + 1×60 + 1 → place-string `1, 1, 1` in base-60 (one  , one  , one).

III Mayan Vigesimal

Base 20 · Vertical Positional

A base-20 system stacked vertically: dots (1) and bars (5) fill each position, and a shell glyph marks zero — one of humanity's earliest written zeros. Each step up the stack multiplies by twenty.

GLYPH	VALUE	NAME	MEANING
	0	Shell (0)	The void, the placeholder — zero as a written concept
●	1	Dot (1)	Single dot — one unit
—	5	Bar (5)	Horizontal bar — five units, the hand
● ● ● ● — — —	9	Nine	Four dots over one bar — the Bolontiku underworld lords
● ● ● ● — — — —	19	Nineteen	Four dots over three bars — max single-position value
↑	20	Twenty	Positional step — triggers the next vertical layer (20^1)
↑ ↑	400	Four Hundred	Twenty squared — two layers up (20^2)
↑ ↑ ↑	8,000	Eight Thou-sand	Twenty cubed — the great calendrical cycle (20^3)

IV Khoisan Click Phonemes

Multi-Channel · Parallel

Not a number system but a parallel encoding: a single click carries 4–6 simultaneous channels of information — tongue position, airflow direction, contact placement, accompaniment and nasalization. One sound = several data streams at once.

C

Dental Click IPA |

Tip of tongue against front teeth, quickly withdrawn

Tongue position · Ingress airflow · Dental placement

Tc

Alveolar-Palatal Click IPA ‡

Tip of tongue against alveolar ridge and adjacent palate, released sharply downwards

Alveolar contact · Palatal friction · Sharp release

Q

Palatal Click IPA !

Tongue pressed against upper palate, released sharply — like a cork pulled from a bottle

Full palate contact · Pressure build · Pop release

X

Lateral Click IPA ||

Click at the side of tongue held pressed against palate

Lateral tongue · Side airflow · Palate slide

V The Glyph Library

The full inscribed set spanning numerals, deities, determinatives and phonetic signs — the raw vocabulary from which Hiero-Code rune nodes draw their pictographic identity.

Egyptian — 17 glyphs

GLYPH	NAME	VALUE	CAT-EGORY	MEANING
𓏺	Staff (1)	1	NUMERAL	Single unit, the tally mark
𓎆	Heel Bone (10)	10	NUMERAL	The first grouping, ten strokes
𓍢	Coil of Rope (100)	100	NUMERAL	The coil, one hundred
𓆼	Lotus Flower (1000)	1,000	NUMERAL	Abundance, the lotus in bloom
𓂭	Bent Finger (10000)	10,000	NUMERAL	Power of the pharaoh's hand
𓆐	Tadpole (100000)	100,000	NUMERAL	The infinite multiplier
𓁨	Kneeling God (1000000)	1,000,000	NUMERAL	The god of eternity, a million
𓋴	Nefr (Zero/Balance)	0	SPECIAL	Beautiful-perfect, balance, the base
𓂀	Eye of Horus	1	SYMBOL	Vision, protection — the Wedjat fractions 1/2…1/64
𓇳	Sun Disk (Ra)	1	DEITY	Solar identity, the source of all light
𓊽	Djed Pillar	1	SYMBOL	Stability, endurance, structural integrity
𓋹	Ankh	1	SYMBOL	Life force, the key of the Nile
𓌀	Was Sceptre	1	SYMBOL	Power and dominion
𓅞	Thoth (Ibis)	1	DEITY	Wisdom, mathematics, writing
𓀀	Man (Seated)	1	FIGURE	Unity, primal form, the first count
𓆑	Horned Viper	10	ANIMAL	Transformation, the tenth power
𓂋	Mouth (r)	1	PHONET-IC	Speech, the glyph of one, output

Sumerian — 17 glyphs

GLYPH	NAME	VALUE	CATEGORY	MEANING
𒁹	Ash (1)	1	NUMERAL	The single vertical wedge — the base unit
𒌋	U (10)	10	NUMERAL	Ten units in sexagesimal notation
𒐐	Géš (60)	60	NUMERAL	One sixty — the base unit of the cosmos
𒐖	Géš-U (600)	600	NUMERAL	Ten sixties — the administrative unit
𒐕	Šár (3600)	3,600	NUMERAL	The great cycle — 60^2, one shar
𒐊	Niš (20)	20	NUMERAL	Two tens, a score
𒐋	Ušu (30)	30	NUMERAL	Three tens, the lunar month
𒐌	Nimin (40)	40	NUMERAL	Two twenties

GLYPH	NAME	VALUE	CATEGORY	MEANING
	Ninnu (50)	50	NUMERAL	Fifty, close to the great base
	Aš (Whole)	1	COUNTING	The complete unit
	Min (Two)	2	COUNTING	Duality, paired forces
	Eš (Three)	3	COUNTING	Trinity, the triad of existence
	Ia (Five)	5	COUNTING	The hand, five fingers
	Dingir (God)	1	DETERMINATIVE	Divinity, the first determinative
	Giš (Tree/ Sixty)	60	DETERMINATIVE	The great wood, also sixty — structural support
	Enki (Wisdom)	40	DEITY	God of wisdom, water, magic — 40 in the pantheon
	Enlil (Air)	50	DEITY	God of wind and storm — 50 in the pantheon

Mayan — 8 glyphs

GLYPH	NAME	VALUE	CATEGORY	MEANING
•	Mayan Dot (1)	1	NUMERAL	Single dot — one unit in base-20
—	Mayan Bar (5)	5	NUMERAL	Horizontal bar — five units, the hand
	Mayan Shell (0)	0	NUMERAL	The shell glyph — zero, the void, the placeholder
• • • • — — —	Mayan 9	9	NUMERAL	Four dots + bar — the Bolontiku underworld lords
• • • • — — — —	Mayan 19	19	NUMERAL	Four dots + three bars — max single-position value
↑	Mayan 20 (Next Layer)	20	NUMERAL	Positional step — triggers the next vertical layer
↑ ↑	Mayan 400	400	NUMERAL	Twenty squared — two positional layers up
↑ ↑ ↑	Mayan 8000	8,000	NUMERAL	Twenty cubed — the great calendrical cycle

VI The Pyri System

The EMOJI Layer · Valence × Arousal × Gesture

Where the older scripts count quantity, Pyri encodes affect. It is the keyboard-native emoji of Hiero-Code: QUERTY-typeable emoticons that serve as macro-nodes for emotional state in machine-learning embeddings.

Why an emoji system belongs beside the numerals

Every other script in this codex answers the question “how many?” The Pyri system answers a different one — “how does the signal feel?” Yet it is built from the exact same architectural instinct that makes the numeral systems powerful: **a tiny alphabet of primitives, combined by rule, spanning an enormous expressive space.** Egyptian repeats seven pictographs; Sumerian positions a handful of wedges across powers of sixty; Pyri takes **12 eye-forms × 8 mouth-forms × 8 gestures** and multiplies them into **768 distinct macro-expressions.** The emoji is not decoration bolted onto a symbolic language — it is the same combinatorial engine pointed at emotion instead of number.

The Three Generative Axes

Eyes — 12 forms

EYE	VALENCE	AROUSAL	READS AS
`>.<`	Negative	High	Frustration / embarrassment
`^.^`	Positive	Low	Happiness / satisfaction
`0.0`	Neutral	Medium	Shock / blank stare
`@.@`	Neutral	High	Overstimulation / dizzy
`*_*`	Positive	High	Awe / love-struck
`$.$`	Positive	Medium	Greed / motivation
`-.-`	Negative	Low	Boredom / deadpan
`~.~`	Neutral	Low	Sleepy / indifference
`U.U`	Negative	Low	Defeat / resignation
`X_X`	Negative	Medium	Done / defeated
`3_3`	Neutral	Low	Dreamy / indifferent
`+.+`	Positive	Medium	Focus / interest

Mouths — 8 forms

MOUTH	SHAPE	VALENCE
.	dot	Neutral
_	flat	Neutral
3	open	Positive
O	surprised	Neutral
V	sad	Negative
W	wavy	Positive
P	tongue	Positive
~	wavy-alt	Neutral

Gestures — 8 forms

GLYPH	GESTURE
∅	None
\(-.-)/	Shrug / Surrender
\(^.^)/	Waving / Joy
(T_T)	Crying
\(O_O)/	Shock
\(*_*)/	Awe
\(>.<)/	Frustration gesture
;;	Sweat drops

The Emotion Vector — how a face becomes a number

Each composed emoji collapses to a three-component vector `[valence, arousal, gesture]`. Eyes contribute valence and arousal; the mouth nudges valence; the gesture adds a motor / embodiment code. Valence and arousal together are the classic circumplex model of affect — the same two axes psychologists use to map every human feeling onto a plane. Pyri adds a third, bodily axis so a node can express not just what it feels but what it is doing about it.

Canonical Emotion Vectors

`*_*`

High Arousal Positive

V +1 | A +1 | G 0

Awe / love-struck — strong engagement modeling

`^.^`

Low Arousal Positive

V +1 | A -1 | G 0

Contentment / pleasure — reward signal

`>.<`

High Arousal Negative

V -1 | A +1 | G 0

Frustration — mild distress node

`U.U`

Low Arousal Negative

V -1 | A -1 | G 0

Defeat / resignation — passive state

`@.@`

Neutral Overstimulation

V 0 | A +1 | G 2

Sensory overload — trigger scanning node

`3_3`

Neutral Dormant

V 0 | A -1 | G 0

Fatigue / passive — low-energy node

`$.$`

Motivated

V +1 | A 0 | G 1

Incentive / motivation — goal-seeking node

`\(-.-)/`

Surrender Gesture

V -1 | A 0 | G 3

Motor-expression mapping — gestural node

Seeded Pyri Glyph Nodes

GLYPH	NODE	V	A	G	MEANING
`>.<`	Frustration Node	-1	+1	0	High arousal, negative valence. Frustration / embarrassment.
`^.^`	Contentment Node	+1	-1	0	Low arousal, positive valence. Happiness / satisfaction.
`0.0`	Shock Node	0	+1	0	Medium-high arousal, neutral valence. Attention / focus trigger.
`@.@`	Overload Node	0	+2	2	High arousal, neutral-negative. Sensory overload detection.
`*_*`	Awe Node	+1	+2	0	High arousal, positive. Strong engagement / awe.
`$.$`	Motivation Node	+1	0	1	Medium arousal, positive. Incentive / goal-seeking drive.
`U.U`	Defeat Node	-1	-1	0	Low arousal, negative. Resignation / passive state.
`\ (-.-)/`	Surrender Node	-1	0	3	Medium arousal, negative gesture. Motor-expression mapping.

Connective Commentary — the emoji system in conversation with the others

↔ Sumerian — place value becomes axis value

Sumerian gets its reach from positional encoding: the same wedge means 1, 60, or 3,600 depending on where it sits. Pyri performs the identical trick across semantic positions instead of magnitudes. A symbol's meaning depends on which slot a primitive occupies — eye-slot, mouth-slot, or gesture-slot. Replace base-60 place value with a three-axis product space and you have the Pyri tensor: `Eyes ⊗ Mouths ⊗ Gestures`. Both systems buy a vast vocabulary with a handful of signs by letting position carry meaning.

↔ Egyptian — additive sentiment

Egyptian numerals simply add up: stack the pictographs and sum them. Pyri's valence works the same way — the eye's valence and the mouth's valence are summed and clamped to Positive / Neutral / Negative. A high-valence eye (^.^) plus a sad mouth (v) cancel toward Neutral, exactly as Egyptian symbols accumulate toward a total. Emotion, here, is arithmetic.

↔ Khoisan — parallel channels of feeling

The Khoisan click is the deepest cousin of the emoji. A single click fires 4–6 information channels simultaneously; a single Pyri glyph fires valence, arousal and gesture simultaneously. Neither is a linear stream of tokens — both pack multiple orthogonal dimensions into one indivisible mark. The emoji is, quite literally, a visual click: one glyph, many parallel readings.

↔ Mayan — a zero for emotion

The Mayan shell gave the world a written nothing — a placeholder that makes higher positions legible. Pyri keeps its own zero: the Neutral valence and the empty gesture (⌀). Neutral is not absence of data; like the shell, it is a meaningful placeholder that lets the surrounding signal be read precisely.

↔ The Rune Engine — affect as a control signal

Hiero-Code rune nodes already carry two continuous scalars: ASE vitality (influence) and Maat balance (stability). The Pyri vector slots straight into that machinery — arousal behaves like a vitality gain, valence like a balance bias, and the gesture code like a discrete action selector. The emoji layer is therefore the **human-readable face of the same numbers the engine already computes**: it makes a model's internal emotional state legible at a glance, using the one script every modern keyboard can already type.

In the Hiero-Code cosmology the ancients counted the heavens in tens, sixties and twenties; the Pyri system counts the inner weather of a mind. Same instinct, new axis — a pictographic alphabet of feeling, 768 expressions deep.

VII The Lineage of Feeling

Emoji Ancestry · 1982 – 2010

The Pyri layer did not appear from nowhere. It is the heir of three real traditions of typed feeling — Western emoticons, Japanese kaomoji, and the carrier emoji of the late-1990s and 2000s. This chapter is the historical reference behind the system.

Two keyboards, one instinct

Long before color emoji, people built faces from punctuation. In 1982 Scott Fahlman proposed `:-)` and `:-(` on a Carnegie Mellon board; Japan answered in 1986 with `(^_^)`. Two grammars emerged: the West reads sideways and weights the mouth; Japan reads face-on and weights the eyes. Pyri is deliberately the second kind — its twelve **eye-forms** carry the affect while the mouth only nudges it, exactly as kaomoji do.

Western emoticons — read sideways

EMOTICON	MEANING	PYRI ECHO
`:-)`	Smile / happy	`^.^`
`:-D`	Grin / laughing	`0.0`
`;-)`	Wink / joking	`+.+`
`:-(`	Sad / frown	`U.U`
`:-P`	Tongue out / playful	`+.+`
`:-O`	Surprised / shocked	`0.0`
`:-/`	Skeptical / unsure	`-.-`
`:'(`	Crying / weeping	`U.U`
`>:-)`	Evil grin / mischief	`>.<`
`B-)`	Cool / sunglasses	`$.$`
`:-|`	Neutral / deadpan	`~.~`
`:-*`	Kiss	`*_*`
`<3`	Heart / love	`*_*`
`</3`	Broken heart	`U.U`
`XD`	Laughing hard	`>.<`
`:-$`	Embarrassed / blushing	`@.@`
`%-)`	Dizzy / overloaded	`@.@`
`o:-)`	Angel / innocent	`^.^`

Kaomoji — read face-on (eye-weighted)

KAOMOJI	MEANING	PYRI ECHO
`(^_^)`	Happy / content	`^.^`
`(>_<)`	Frustration / wincing	`>.<`
`(T_T)`	Crying	`>.<`
`(;_;)`	Quiet sobbing	`U.U`
`(._.)`	Blank / neutral	`~.~`
`(=_=)`	Exhausted / deadpan	`-.-`
`(¬ω¬)`	Side-eye / unimpressed	`3_3`
`(♡ω♡)`	In love / adoring	`*_*`
`¯\_(•_•)_/¯`	Shrug / “I dunno”	`\ (-.-)/`
`(╯°□°)╯ ~ ┻┻`	Table flip / rage	`>.<`
`m(_ _)m`	Deep bow / apology	`U.U`
`\(^o^)/`	Celebration / joy	`\ (^.^)/`

The Carrier Emoji Era — Japan, 1997 onward

The pictographic emoji proper was born on Japanese mobile phones, where a 250-character limit forced designers to pack meaning into 12×12-pixel icons. Three carriers each shipped their own mutually-incompatible set — the original reason a single code could mean a convenience store on one network and a wristwatch on another, a problem later resolved by folding emoji into the Unicode standard.

1997	**J-Phone (later SoftBank)** 90 emoji on the SkyWalker DP-211SW — 12×12 px, monochrome. The first mobile emoji set, including an early pile-of-poo. Later shaped Apple's emoji.
1999	**NTT DoCoMo** Shigetaka Kurita's 176 emoji for the i-mode service — single-hue on a 12×12 grid. This set was later acquired by the Museum of Modern Art.
1999	**au by KDDI** Type A-2: 170 black-and-white emoji for EZweb. Gained animation in the early 2000s and later seeded Google's emoji set.
2000–03	**Carrier expansion** au Type B-2 (188 emoji), B-3 (added shading), then the first animated sets. Each carrier's emoji stayed mutually incompatible across networks.
2008	**Apple & Google** Apple Color Emoji ships on iPhone OS using SoftBank's private-use scheme; Google adds emoji to Gmail in collaboration with au by KDDI.
2010	**Unicode 6.0** Carrier emoji formally standardized into Unicode — the bridge from the fragmented carrier era to today's universal set.

Inside DoCoMo's Original 176 — Kurita's ten categories, 1999

Shigetaka Kurita's set — now in the Museum of Modern Art — organized its 176 glyphs into ten thematic families. They map cleanly onto the structure Hiero-Code uses today: most are nouns (things and places), but one whole family is pure affect — the direct ancestor of Pyri.

CATEGORY	REPRESENTATIVE CONTENTS
Weather & Nature	Sun, cloud, rain, snow, typhoon, moon, sakura, autumn leaf
Faces & Emotions	Happy, sad, angry, surprised, sleepy, sick, love-struck, cool
Hearts & Symbols	Heart, broken heart, the card suits ♠ ♥ ♦ ♣, question mark
Communication & Tech	Mobile phone, telephone ☎, pager, TV, computer, envelope ✉
Transport & Places	Car, taxi, bus, train, airplane, ship, house, hospital
Food & Drink	Noodles, bento, rice, beer, coffee ☕, cake, fruit, fast food
Sports & Activities	Soccer, baseball, tennis, skiing, music note, microphone, games
Time & Calendar	The twelve clock faces, one o'clock through twelve
Zodiac	The twelve signs ♈ ♉ ♊ ♋ ♌ ♍ ♎ ♏ ♐ ♑ ♒ ♓
Money & Business	Yen, rising chart, 100-points, OK / NEW / UP / FREE markers, recycle ♲

↔ Why this belongs in the Codex

Every ancestor above solved the same problem the numeral systems solve: a tiny set of typeable marks, combined by rule, spanning a huge space of meaning. Western emoticons and kaomoji proved a face could be built from keyboard primitives; the carrier sets proved those

> faces could be standardized and counted. Pyri inherits both lessons — it is a face-on, eye-weighted, QUERTY-typeable alphabet of emotion, descended in a straight line from `(^_^)`.

From *:-)* in 1982 to a 768-expression tensor — the same instinct, carried forward one keyboard at a time.

VIII Iñupiaq · Arctic Indigenous

Base-20 · sub-base 5

The counting language of the North Slope Iñupiat — a vigesimal system built on the human body: five fingers, a hand, and finally iñuiññaq, "a whole person" of twenty digits. A living lexicon of several thousand entries underpins this chapter.

Iñupiaq numbers count in twenties with a five-based interior: six is itchaksrat, seven is literally "five-two", and nine is "lacking one to ten". Twenty is a whole person; forty, sixty and eighty are simply two, three and four persons. This body-grounded base-20 is the same arithmetic skeleton the Maya and Aztec share — counted on a different continent, under the aurora rather than the sun.

Vigesimal Numbers

VALUE	IÑUPIAQ	LITERAL
1	**atausiq**	—
2	**malġuk**	—
3	**piŋasut**	—
4	**sisamat**	—
5	**tallimat**	—
6	**itchaksrat**	—
7	**tallimat malġuk**	five-two
8	**tallimat piŋasut**	five-three
9	**quliŋŋutaiḷaq**	lacking one to ten
10	**qulit**	—
15	**akimiaq**	—
20	**iñuiññaq**	a whole person (20 digits)
40	**malġukipiaq**	two persons
60	**piŋasukipiaq**	three persons
80	**sisamakipiaq**	four persons
100	**tallimakipiaq**	five persons
400	**iñuiññaq iñuiññaił**	twenty twenties

Colour Terms

IÑUPIAQ	ENGLISH
qauluktaq	black
qatiqtaq	white
aupallaktaq	red
quŋŋuqtaq	blue / green
suŋaaqtaq	yellow
tunŋuqtaq	dark / blue-black
quġžuqtaq	grey

> In the dictionary layer, each entry carries its dialect tags (e.g. North Slope, Kobuk) so a single concept can be read across the Arctic's regional variants — a parallel, multi-channel reading that echoes the Khoisan click.

Writing-Key Phonemes

The practical orthography adds letters English lacks — uvular stops, lateral fricatives, and nasals — each a distinct articulatory "channel". These are the glyph primitives the Iñupiaq rune nodes draw upon.

ġ

gamma

Voiced velar fricative (back 'g')

ɬ

barred-L

Voiceless lateral fricative

ɻ

retroflex L

Retroflex lateral approximant

ñ

enye

Palatal nasal (as in 'canyon')

ŋ

eng

Velar nasal (as in 'sing')

q

uvular Q

Voiceless uvular stop (deep 'k')

ż

z-dot

Voiced retroflex fricative

k

kak

Voiceless velar stop

↔ Into the Rune Engine

Two Iñupiaq-inspired procedure codes join the engine: `IGLU_INSULATE`, which damps a node's ASE volatility the way snow-block walls hold heat, and `SINIK_DORMANT`, a winter-sleep state that parks a node until re-woken — survival logic, rendered as compute.

IX The Aztec Calendar

Tonalpohualli · 260 + Xiuhpohualli · 365

The Mexica reckoned time as two meshing wheels: a 260-day sacred count of twenty day-signs against thirteen numbers, and a 365-day solar year of eighteen twenty-day months plus five hollow days. After David Stuart, The Aztec Calendar Stone: King and Cosmos.

The Twenty Day-Signs · Tonalpohualli

Each day pairs one of these twenty signs with a number 1–13; the two cycles realign only after 260 days. The signs run from Cipactli (the primordial crocodile-earth) to Xochitl (flower), a full arc of creation.



1. Cipactli

Crocodile — Primordial earth-monster; genesis, origin



2. Ehecatl

Wind — Breath of Quetzalcoatl; spirit, change



3. Calli

House — Shelter, rest, the western quarter



4. Cuetzpalin

Lizard — Abundance, regeneration, sun-warmth



5. Coatl

Serpent — Life-energy, the eastern quarter



6. Miquiztli

Death — Transformation, the lord Mictlantecuhtli



7. Mazatl

Deer — Grace, the hunt, sustenance



8. Tochtli

Rabbit — Fertility, pulque, year-bearer of the south



9. Atl

Water — Flow, the flood that ended the 4th Sun



10. Itzcuintli

Dog — Guide of the dead through Mictlan



11. Ozomatli

Monkey — Play, art; the people of the 2nd Sun



12. Malinalli

Grass — Endurance, the twisting of fibres



13. Acatl

Reed — Authority, year-bearer of the east



14. Ocelotl

Jaguar — Night, earth, the devourers of the 1st Sun



15. Cuauhtli

Eagle — Sky, the warrior, the solar ascent



16. Cozcacuauhtli

Vulture — Long life, wisdom of age



17. Ollin

Movement — Earthquake; sign of the present Fifth Sun



18. Tecpatl

Flint — Sacrifice, the obsidian blade, year-bearer north



19. Quiahuitl

Rain — Tlaloc; the fire-rain that ended the 3rd Sun



20. Xochitl

Flower — Beauty, song, completion of the count

Vigesimal Place-Values

VALUE	GLYPH	NAHUATL	SIGN
1	•	**Ce**	Dot (unit)
20		**Cempohu-alli**	Pantli (flag)
400		**Centzontli**	Tzontli (feath-er/hair)
8,000		**Cenxiqui-pilli**	Xiquipilli (in-cense bag)

> Aztec arithmetic is base-20: a dot is one, a flag (pantli) twenty, a feather (tzontli) four hundred, an incense-bag (xiquipilli) eight thousand. The engine's `SCALE_20` and `SUB_BASE_5` codes mirror this directly.

The Four Year-Bearers

NAHUATL	ENGLISH	DIRECTION
Acatl	Reed	East
Tecpatl	Flint	North
Calli	House	West
Tochtli	Rabbit	South

Only four day-signs can open a solar year, each tied to a cardinal direction — the calendar's compass. The 365-day xiuhpohualli is eighteen veintenas of twenty days plus the five nameless nemontemi — hollow, unlucky days that close the ring.

The Five Suns

Mexica cosmology counts five world-ages, each a "Sun" destroyed in turn. We live in the Fifth, Nahui Ollin — 4 Movement — fated to end in earthquakes, the day-sign Ollin at the stone's centre.

SUN	NAME	READING	DESTRUCTION
1	**Nahui Ocelotl**	4 Jaguar	Inhabitants devoured by jaguars
2	**Nahui Ehecatl**	4 Wind	Swept away by hurricanes; people became monkeys
3	**Nahui Quiahuitl**	4 Rain	Consumed by a rain of fire; people became turkeys
4	**Nahui Atl**	4 Water	Drowned in a great flood; people became fish
5	**Nahui Ollin**	4 Movement	The present Sun, fated to end in earthquakes

↔ Into the Rune Engine

Two calendar codes drive the engine: `TONALPOHUALLI_CYCLE` wraps a node's value on the 260-count (so 300 folds to 40), and `NAHUI_OLLIN` applies the "4 Movement" transform — a periodic upheaval that re-balances Maat across connected nodes.

Two wheels, twenty signs, five suns — the Aztec turned time itself into a rune that the engine can now turn in kind.

X

British Museum Stelae Corpus

95 catalogue objects · Old Kingdom

A working transcription of Hieroglyphic Texts from Egyptian Stelae &c., the British Museum's plate catalogue of funerary monuments — false doors, offering tables, panels and reliefs from the Third through Sixth Dynasties. Imported best-effort from the original OCR, OCR noise and all, as a live reference layer beside the rune engine.

Each plate records a single monument: its title, British Museum registration number, stone, provenance, dimensions, dynasty, and the curator's description of the carved scene. Of the 95 objects parsed, 63 carry a BM number, 90 a descriptive note, and 85 a recorded provenance — a partial but substantial index into one of the oldest bodies of hieroglyphic writing. The live app exposes the full set as a searchable, filterable corpus; this chapter is its printed core.

By Dynasty

PERIOD	COUNT	SHARE
Third Dynasty	1	1%
Fourth Dynasty	34	35%
Fifth Dynasty	20	21%
Sixth Dynasty	24	25%
Unattributed	16	16%

> The corpus skews to the Fourth and Fifth Dynasties — the great age of the mastaba field at Giza and Saqqara, when private funerary stelae and false doors proliferated.

By Object Type

TYPE	COUNT
Object	26
False Door	19
Slab	15
Stela	6
Panel	6
Relief	5
Fragment	5
Block	3

Common Materials

STONE	COUNT
Limestone	87
Pinkish sandstone	1
imestone	1
Limeston:	1
L nestone	1
mestone	1
Limestor	1

Selected Catalogue

A representative slice of the named monuments carrying a British Museum registration number. Plate numbers follow the original volume; titles and stones are reproduced as transcribed.

PL.	TITLE	BM NO.	MATERIAL	DYNASTY
11	1 and 2. Pancls of Bw-nfr (J.	1274	Limestone	Fourth
11	3. Block from tomb of Re-firp.	1277	Limestone	Fourth
111	1. Slab with figure of Try	1169	Limestone	Fourth
IV	2. Slab with figure of 71 (=	161	Limestone	Fourth
V	3. False door of Ki(:i)1p.	1288	Limestone	Fourth
VIII	2. False door of Was(i)mir(i)(U] 9).	1223	Limestone	Fifth
VIII	3. Slab of Rudk(:1)	1269	Limestone	Fourth
VIII	4. Door-drum of RudA(:).	1268	Limestone	Fourth
IX	1. False door of Hinw (°	1272	Limestone	Fourth
X	2. Block from the tomb of Nfr-Mort(15 =)	1510	Limestone	Fourth
XI	2. Slab with the name of the tomb of Shepseskaf.	1234	Limestone	Fourth
XII	1. Slab from a tomb.	186	Limestone	Fifth
XII	3. Slab from the tomb of Ny-ibu-ni.	1266	Limestone	—
XIII	1. Panel of 27 (1)),	1212	Limestone	Fourth
XIII	2. Stela of Mre-ir (V22 1).	128	Limestone	Fourth
XIII	3. Panel from the tomb of Sri ().	1192	Limestone	Fourth
XVIII	1. Pancl.	1166	Limestone	Fifth
XX	2. Small false door of Renfr (1)	1011	Limestone	Fifth
XXI	1. Architrave of Ny-ki(:)-<nh (5 7).	1275	Limestone	Fifth
XXI	2. Architrave of IWss-Pil (31 £	1278	Limestone	—
XXII	1. Tomb-fragment.	65951	Limestone	—
XXII	2. Offering-table of Hib-n (2)	65955	Limestone	Fifth
XXV	1. Cone of Hiww (or Hir	199	Limestone	Fifth
XXV	2. Fragment of n inscription.	992	Limestone	Fifth
XXV	3. Part ofa scene from 4 tomb.	994	mestone	—
XXVI	1. Panel	14298	—	Fifth

↔ Into the Rune Engine

The corpus grounds the Egyptian rune system in real monuments: every false door and offering formula is a fixed program — a sequence of glyph-nodes encoding the deceased's name, titles, and the ḥtp-dỉ-nsw offering procedure. Read this way, an Old Kingdom stela is the oldest surviving Hiero-Code: a symbolic procedure carved to run for eternity.

Five dynasties of carved stone, indexed and searchable — the corpus turns a museum catalogue into a living dictionary the engine can read.

XI Hindu Iconography

63 entries · sacred image-grammar

A working transcription of a Hindu iconography reference — the symbolic vocabulary of sacred imagery, where every posture, gesture, ornament, and attribute carries fixed meaning. Imported best-effort from the original OCR, noise and all, as a live reference layer beside the rune engine.

Hindu icons are read, not merely seen: the number of arms, the weapon in each hand, the mudrā (hand-gesture), the vehicle (vāhana), and the throne all encode doctrine. Of the 63 entries parsed, 62 carry an explanatory meaning and 39 an English gloss — spanning iconology theory, recurring iconographic features, the canonical hand-gestures, and a glossary of iconic symbols. The live app exposes the full set as a searchable, filterable reference; this chapter is its printed core.

By Category

CATEGORY	COUNT	SHARE
Iconic Symbol	39	61%
Iconology Theory	12	19%
Iconographic Feature	6	9%
Mudra	6	9%

The bulk of the corpus is the iconic-symbol glossary — Sanskrit terms paired with their esoteric meaning — framed by iconology theory and the canonical hand-gestures.

The Mudrās · Canonical Hand-Gestures

Abhaya Mudra

the gesture of fearlessness — the palm displayed with the fingers pointing to the sky. Indicates the practice of harmlessness to all beings in word, d

Varada Mudra

the gesture of generosity — the palm displayed, fingers pointing downwards. This indicates the benevolence of the deity as well as the teaching of the

Cin Mudra

the gesture of teaching — palm displayed, fingers upwards thumb and fore-finger joined at the tips. This indicates the imparting of knowledge, form th

Dhyāna Mudra

the gesture of meditation, this takes two forms: palms placed one over the other in the lap or arms stretched out, hands on the knees with the palms o

Tarjani Mudra

the gesture of vigilance — fist closed with the index finger raised. This mudra is usually found in images of protective or guardian deities and indic

Kaṭi-hasta Mudra

this is found almost exclusively in icons of Lord Vishnu as Venkaṭeśvara and is linked to his other hand in the Varada-Mudra which is the gesture of g

Selected Iconic Symbols

A representative slice of the glossary: each Sanskrit term paired with its English gloss and the symbolic meaning ascribed to it in the source. Reproduced as transcribed.

SYMBOL	ENGLISH	MEANING
Pāśa	Noose	The 3 bonds that bind us to the cycle of rebirth – avidya/māyā (ignorance about the nature of reality) karma (action) vā
Aṅkuśa	Elephant Goad	Incentive to continue in spiritual practice and the application of the teachings. The urging of the guru which drives us
Churi	Dagger	The spiritual path - the razor's edge which needs to be trod with great care and vigilance. The dagger is razor sharp an
Vīnā	Lute	Inner sound of spiritual harmony, the harmony of inner being, the vibration of the atoms in motion, creation, the sound
Kamaṇḍalu	Water jug	Fullness and generosity, the water of life and also purity and purification.
Gadā	Mace	Primarily a symbol of Vishnu but also shared among many of the deities. The mace is a symbol of sovereignty and the Cosm
Śālipaḷḷava	Rice-sprig	The bounty of prosperity, fecundity and abundance. The symbol of nourishment, abundance, and the sacred cycle and contin
Dhvaja	Flag	Making oneself known to others, indicating who you are and what you stand for. An indication of the triumph of the Dharm
Khatvāṅga	Club with skull	Impermanence of all phenomena, dissolution of the elements. Also represents the 8 mystical powers obtained through yoga
Kheṭaka	Shield	Security, defence, protection of oneself and the Dharma, deflection of negativity and assault by others. The shield defi
Ḍamaru	Hour-glass drum	Held in the hand of Lord Śiva. It represents the union of the masculine and feminine principles and the projection of th
Padma	Lotus	By reproducing from its own matrix rather than the soil the lotus is a symbol of spontaneous generation (svayambhu). It
Śaṅkha	Conch	In battle conveys the instruction of the commander to the troops. It represents the diffusion of the Dharma teachings an
Cakra	Discus / wheel	The wheel is the symbol of the Dharma which rotates and spins its beneficial influence in all directions. It also symbol
Sarpa	Snake	Kundalini - sexual energy latent within the lowest chakra – the muladhara at the base of the spine. Also symbolizes the
Śuka	Parrot	Truth, the transmission of the teachings. The parrot repeats exactly what it hears without clarification, modification o
Pātra	Bowl	In the hands of a Rishi or the Buddha it symbolizes the begging bowl and the idea of generosity. In the hands of the wra
Paraśu	Axe	Non-attachment - the severing of our ties and bonds to the material world.

SYMBOL	ENGLISH	MEANING
Ghaṇṭa	Bell	Impermanence. The phenomenal world which is impermanent and evanescent. Creation of the transient universe through sound
Cāmara	Fly-whisk	The obedience to the law and in particular to the highest principle of Dharma — ahimsa. Also as a yak tail, represents t
Vajra	Thunderbolt	Diamond wisdom — the ultimate truth as adamantine and indestructable, but which destroys all that is other than truth. S
Triśūla	Trident	Control over action, speech and thought. Also fire — Agni and its 3 forms. The 3 paths to liberation Bhakti – love, Jñān
Kunta	Spear	One-pointedness of concentration applied during meditation aimed at the goal of perfection. Focussed attention at elimin
Khaḍga	Sword	Sword of wisdom which cuts through illusion and destroys all ignorance (viveka) - hidden within all of us like a scabbar
Pustaka	Book	The Vedas – sacred Scriptures and the formal learning of all sorts of knowledge and theory.
Japa-Mālā	Rosary	Sādhana or spiritual practice. Meditation and the recitation of mantras.
Cāpa / Śara	Bow & arrow	Concentration and focussing of the mind upon the goals of overcoming lower nature and achieving transformation, enlighte
Ādarśa	Mirror	Notion of the evanescence of the material delusion. The world is but a reflection in the mirror of the pure mind. So the
Kalaśa	Vase	The receptacle of water which is the foremost representative of life in general, water is also allied to breath and all-
Mṛga	Deer	The deer symbolises gentleness as well as attentiveness — whatever its doing the deer is always mindful of predators. Th

↔ Into the Rune Engine

A Hindu icon is a fixed program: the deity's identity is the sum of its attribute-nodes — each arm, weapon, and mudrā a glyph with a stable meaning, composed into a readable whole. Read this way, an icon is Hiero-Code carved in bronze and stone: a symbolic procedure whose output is a god, recomputed in the mind of every worshipper who knows how to read it.

A grammar of gesture and attribute, indexed and searchable — the reference turns a devotional vocabulary into a living dictionary the engine can read.

XII Northwest Coast Art

27 chapters · the formline grammar

A working transcription of an overview of Pacific Northwest Native Indian art — the bold black-and-red carving tradition of the Pacific coast, where every creature is built from a small vocabulary of reusable shapes. Imported best-effort from the original OCR, noise and all, as a live reference layer beside the rune engine.

Northwest Coast art is composed, not drawn freehand: a black formline outlines the body, and inside it a fixed set of shapes — the ovoid, the U-form, the S-form — fill the joints, eyes, ears, and feathers. The same grammar renders a raven, a bear, or a killer whale; only the proportions and attributes change. Of the 27 chapters parsed (~9,700 words), the reference spans the basic elements and colors, the shape vocabulary, animal designs, totem poles and masks, authenticity and buying guidance, and profiles of working carvers. The live app exposes the full text as a searchable, filterable reference; this chapter is its printed core.

By Category

CATEGORY	COUNT	SHARE
Overview	7	25%
Art Element	6	22%
Art Form	3	11%
Animal Design	7	25%
Carver Profile	4	14%

The shape vocabulary (Art Elements) and the animal designs are the heart of the tradition — the reusable glyphs and the creatures composed from them — framed by overview chapters and the carver profiles.

The Crest Animals

The Thunder-bird

The thunderbird has been one of the most dominant icons in Native American art and legends.

The Bear

The Native American people have always had a special place in their hearts for the bear.

The Eagle

The eagle has been a universal symbol of power respected by many cultures throughout the world.

The Killer Whale

One of the most awe-some creatures in both real life and in Pacific Northwest Native Indian art is the orca or killer whale.

The Raven

The most important symbol to many Northwest Native people is the Raven bird who is con-sidered the Creator's assistant.

The Salmon

There are several spe-cies of salmon fish in the Pacific Northwest region.

The Shape Vocabulary · Reusable Form Elements

Every subject is assembled from these elements. The formline carries the outline; the ovoid, U-form, and S-form fill and connect the body. Reproduced as transcribed.

ELEMENT	DESCRIPTION
The Colors of Northwest Coast	The main traditional colors of Pacific Northwest Native Indian art are black and red.
The Shapes & Form of Northwest	With further analysis of Pacific Northwest Native Indian art, it can be realized that the different shapes used by Northwest Nativ
The Formline	The formline of Pacific Northwest Native Indian art is the main line that outlines the body of the subject whether it is a person
The Ovoid Shape	The ovoid is the most dominant shape in Pacific Northwest Native Indian art.
The U-Form Shape	The second most common shape used in Pacific Northwest Native Indian art is the U-form.
The S-Form Shape	Another shape used by Northwest Native artists is the S-form.

The Carvers

CARVER	PROFILE
Gary Baker	It's the eyes that make Pacific Northwest Native Indian art carver Gary Baker unique among his peers.
Peter Charlie	Born in 1957, Peter Charlie of the Squamish Nation in North Vancouver, Canada learned to carve from his grandfather at t
Cody Mathias	Saying that Cody Mathias comes from a large family is an understatement.
Paul Joseph	One of the highlights of Paul Joseph's career so far was to meet the King and Queen of Spain for whom he made a totem po

↔ Into the Rune Engine

A formline design is a fixed program: a creature is the sum of its shape-nodes — each ovoid, U-form, and S-form a glyph with a stable role, composed into a readable whole. Read this way, a carved bear or raven is Hiero-Code worked in cedar and paint: a symbolic procedure whose output is an animal, recomputed in the eye of every viewer who knows how to read the line.

A grammar of shape and crest, indexed and searchable — the reference turns a coastal carving tradition into a living dictionary the engine can read.

XIII Inuit Symbology

22 entries · nalunaikutanga

A working transcription of Nelson Graburn's study of nalunaikutanga — "distinctive features," the signs by which Canadian Inuit recognize animals, places, and people, and which carry over into carving and printmaking. Imported best-effort from the original OCR, noise and all, as a live reference layer beside the rune engine.

In Inuit thought a thing is known by its nalunaikutaq — literally its "de-confuser," the single feature that tells it apart from everything it might be mistaken for: the polar bear by its gait, the walrus by its tusks, a person by name and parka. Of the 22 entries parsed (~4,891 words), the reference spans the essay's analytical sections — semantic structure, the context of recognition, contemporary Inuit arts — and a catalogue of the distinctive features themselves, from nanuk the bear to the inuksuk and the carver's signature. The live app exposes the full text as a searchable, filterable reference; this chapter is its printed core.

By Category

CATEGORY	COUNT	SHARE
Section	8	36%
Distinctive Feature	14	63%

The analytical **Sections** frame the theory of recognition; the **Distinctive Features** are the working vocabulary — the recognition-marks of the animals, people, and places the Inuit carve and print.

The Distinctive Features

Polar Bear · nanuk

The polar bear (nanuk) is recognized by its incisors (tulluriak) and claws (kuki), suggesting its striking power, as well as its tongue (oqak).

Walrus · aivik

The walrus (aivik), the polar bear's only serious opponent, is known mainly by its tusks (tugak) and its moustaches (umik).

Caribou · tuktu

Caribou are known by their antlers (najjuk) and their hooves (kuki), which leave very distinctive footprints (tumit) in the snow.

Owl · uppik

The owl (uppik) is perhaps the most distinctive bird of prey, known by its hooked beak (sigguk), striking claws (kuki), and round, forward-facing eyes

Raven · tulugak

The raven (tulugak) is known by its black feathers; traditional Inuit mythology takes pains to explain the colours of such birds.

Canada Goose · nirlik

Canada geese (nirlit) were traditionally known by their neck markings and calls, and nowadays additionally by the rings that encircle their legs.

Narwhal · aglunguaq

The narwhal (aglunguaq) exhibits its tusk (tugak) as its prime distinctive feature.

Beluga · qilaluga

The beluga (qilaluga) is known by its white skin (makuik).

Common Seal · natsiq

The common seal (natsiq) is known by its inquisitive face and the rings on its pelt; the square-flipper seal by its beard.

Women · the amauti

The bowl-like ovoid opening of the amauti — the woman's parka with its large back pouch for carrying a child — is perhaps the prime distinctive featur

Men · weapons & the kayak

Adult men are recognized by their knives, harpoons, and kayaks — the hunting weapons that distinguish male role and activity.

Inuksuk · the stone marker

Where the land bears no natural markers, the Inuit build characteristic piles of stones — inuksuk, 'imitation person' — in visible places to mark rout

Names · atiit

Names (atiit) act as distinctive features when they refer to an observable characteristic — a scar, stature, a limp, a squint — though birth-names are

The Artist's Signature

Since the late 1950s the signature is the most obvious and widespread form of individual artistic nalunaikutanga, marking who made a work rather than

The Essay · Analytical Sections

Graburn's argument moves from a formal account of the sign to the social settings in which recognition happens, and on to the contemporary art that trades on it. Reproduced as transcribed.

SECTION	OPENING
Summary	Summary: The caneept na]unaikulak is applied in many contexts in Eskimo life anel i s best glossed in English as: sign, symbol, marker, guid
Zusammenfassung (German Summary)	Zusammenfassung: Der Begriff nahmaikutak (Sing.) wird im Leben der Eskimo (Inuit) in vielen Zus amm enh an qen angewandt.
Introduction	In t rod u c t ion 1) Contemporary Inuit commercial arts have been with us for more than a quarter of a century and have been subject to ext
Semantic Structure	The word nalunaikutanga ") is heard in many spheres of Eskimo life with the qross semantic content which may be glossed in English as: sign,
The Context of Recognition	Let us now consider the range of applications of the concept nalwwilwtanga starting with the widest and most natural and ending wih the most
Nalunaikutanga in Contemporary I	Nalunaikutanga enter into the process of Inuit arts in two ways.
Discussion and Conclusions	The Central Canadian Inuit are "interested not in scenery but in action" (Carpenter, 1973: 41).
References	Cited Ca r pe TI te r , E.

↔ Into the Rune Engine

A nalunaikutaq is a recognizer: the minimal feature that resolves an ambiguous input to a single identity. Read this way, every distinctive feature is a Hiero-Code classifier node — a glyph that takes the world's noise and returns one name. The Inuit hunter reading a far silhouette on the ice and the engine resolving a symbol to its meaning run the same procedure: de-confuse, then act.

A semiotics of survival, indexed and searchable — the reference turns a culture's art of recognition into a living dictionary the engine can read.